

RHOAI API Contract Initiative

Stakeholder brief · Date: October 1, 2026

Status: in review

Move API compatibility feedback into component pull requests so teams can catch integration failures earlier, with clear ownership and a repeatable path for intentional changes.

Why this matters

The August 2026 Dashboard gap analysis classified **132 of 194 reviewed issues as upstream component bugs** and found no consumer-facing contract tests in the ten assessed repositories. Dashboard integration tests frequently reveal provider failures after components are assembled, creating rework and release risk. These findings motivate earlier checks; they do not mean every historical bug would be prevented by contract testing.

Proposed approach

Provider teams keep canonical API definitions and behavior tests beside their implementation. Consumers, including Dashboard, record the operations and behavior they depend on. The API Contract testing infra supplies shared compatibility policy, reusable CI checks, and a catalog pointing to these artifacts.

Each provider pull request generates a fresh contract, checks that committed artifacts are current, and compares against protected branch and supported-release baselines. Consumer expectations and provider behavior tests cover semantics that schema comparisons cannot prove. Checks begin in report-only mode; enforcement follows evidence and owner agreement. Intentional breaks require a reviewed migration or deprecation plan and affected-consumer approval. Cross-component release conformance is part of the longer-term model.

First pilot: Model Catalog → Dashboard

The proposed **October 5–16, 2026** pilot covers the Model Catalog REST v1 API used by the Dashboard Model Catalog BFF. It will add a report-only OpenAPI compatibility check and provider-facing consumer coverage for model/version list and lookup, pagination, not-found, and authorization behavior.

Week 1: confirm owners and baselines; integrate the provider check and consumer profile.

Week 2: exercise compatible, breaking, and stale-artifact changes; publish a scorecard and recommend whether to enforce or continue reporting.

Operator/CRD checks and a multi-component release matrix are follow-on work. The broader cohort includes opendatahub-operator and workbenches-operator.

Ownership and stakeholder support

Responsibility	Proposed representative / commitment
Facilitation	Edson Tirelli
Consumer profile and tests	Anthony Coughlin / Dashboard
Provider implementation and checks	Edson Tirelli / Model Catalog
Supported-release baseline	Rishab Prasad / Radim Kubis; confirm provider revision

Success and decision requested

Success means seeded incompatibilities and stale artifacts are detected, an additive change passes, feedback arrives in **under five minutes without a cluster**, results are reproducible locally, and at least one real pull request runs without an unexplained false block. Deliverables are a scorecard, reusable onboarding template, and an enforce/continue-reporting recommendation.

Tracking: [RHOAIENG-96904](#) · **Basis:** Architecture proposal, pilot definition, and gap analysis.